

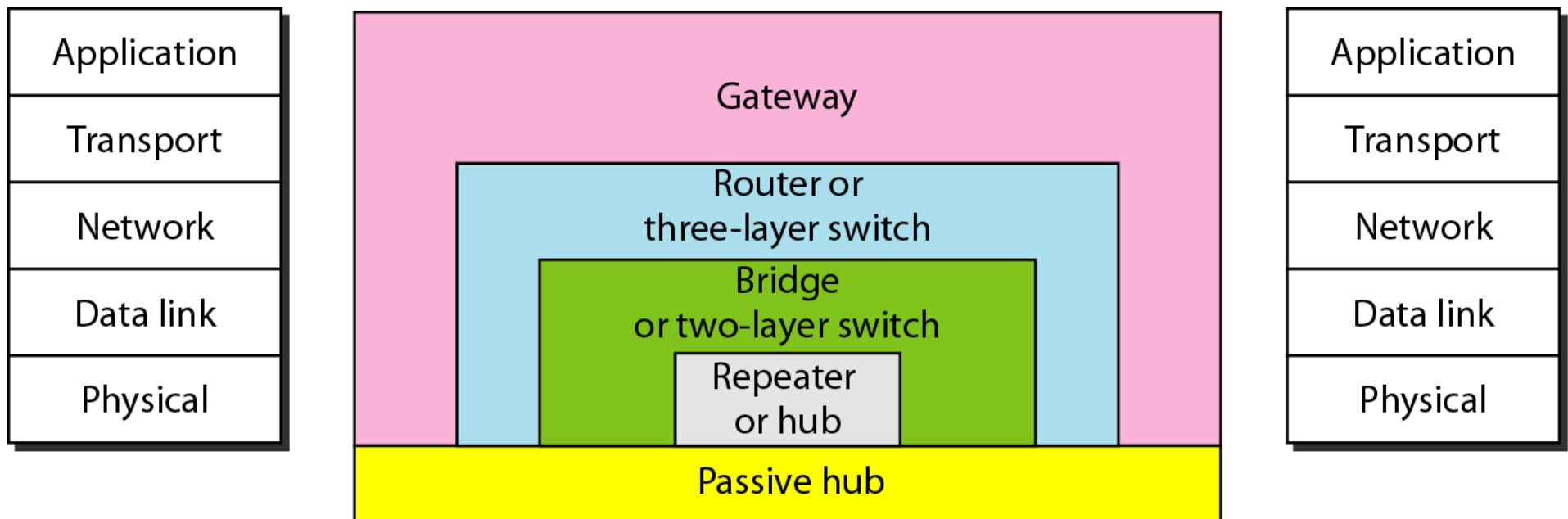


Backbone Networks and Connecting devices

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Connecting Devices



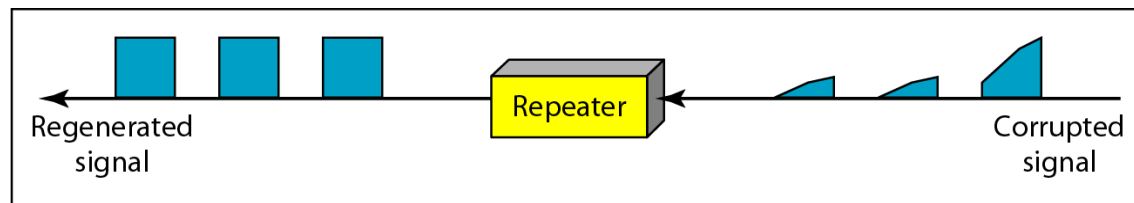


Passive Hub and Repeater

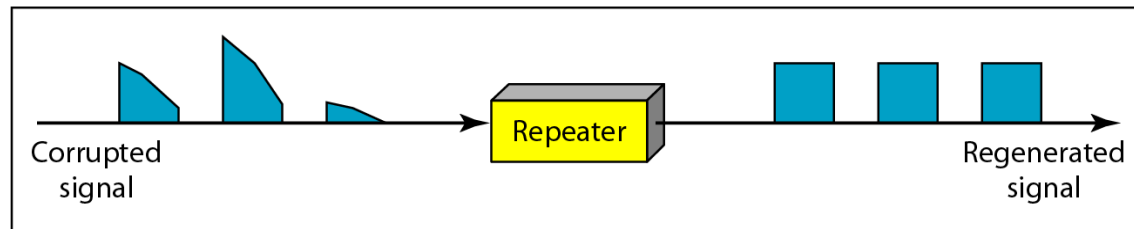
- **Passive Hub**
 - Just a connector
 - It connects the wires coming from different branches.
 - This type of a hub is part of the media; its location in the Internet model is below the physical layer
- **Repeaters**
 - Operates only in the physical layer
 - A repeater receives a signal and, before it becomes too weak or corrupted, regenerates the original bit pattern
 - The repeater then sends the refreshed signal
 - It can extend the physical length of a LAN
 - **A repeater connects segments of a LAN**

Repeater

- A repeater can overcome the 10Base5 Ethernet length restriction
- A repeater forwards every frame; it has no filtering capability
- A repeater is a regenerator, not an amplifier
- A repeater must be placed so that a signal reaches it before any noise changes the meaning of any of its bits



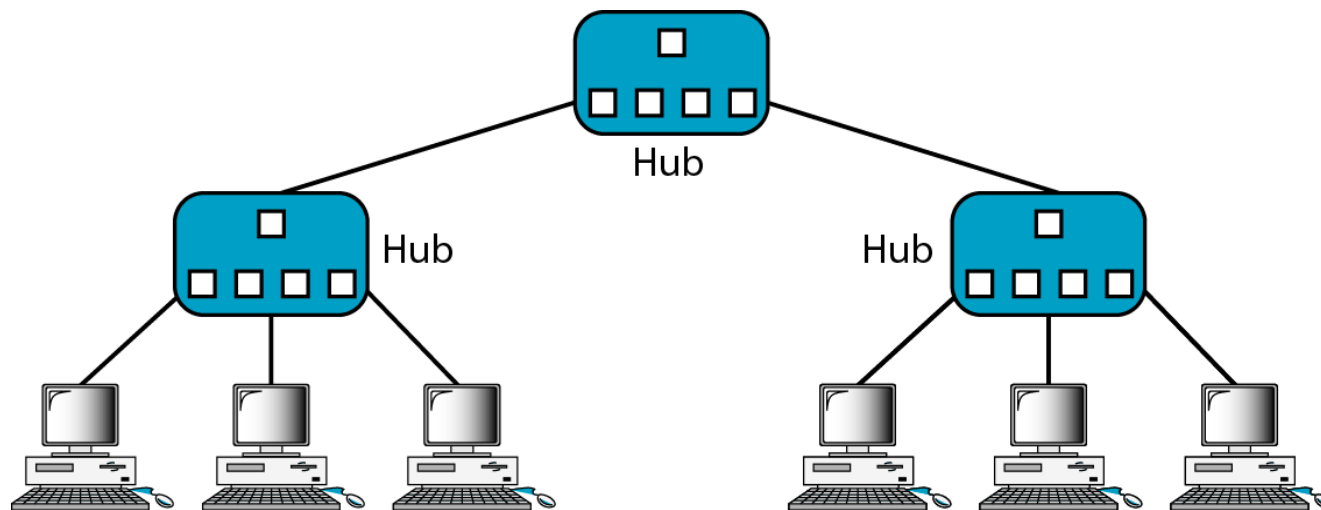
a. Right-to-left transmission.



b. Left-to-right transmission.

Active Hubs

- An active hub is actually a multipart repeater
- Normally used to create connections between stations in a physical star topology
- Hubs can also be used to create multiple levels of hierarchy

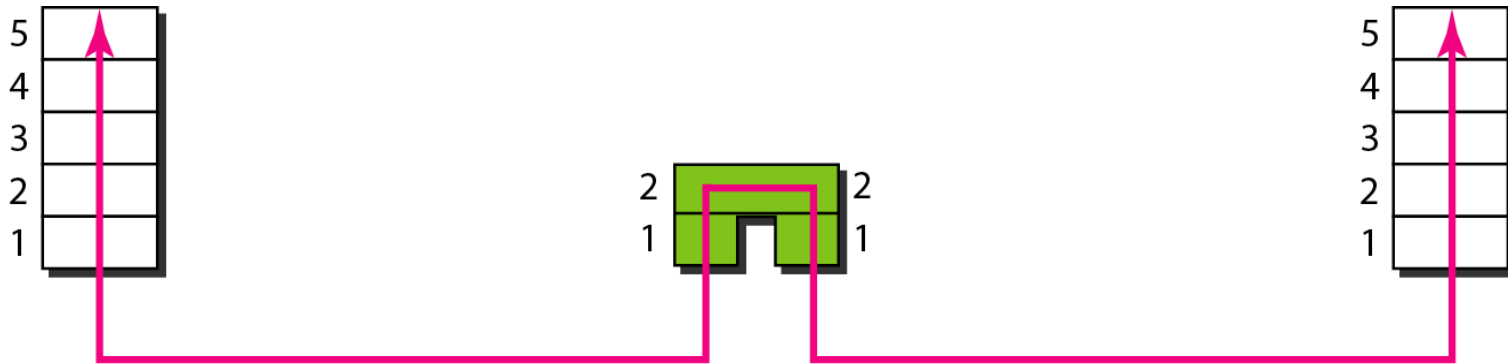




Bridge

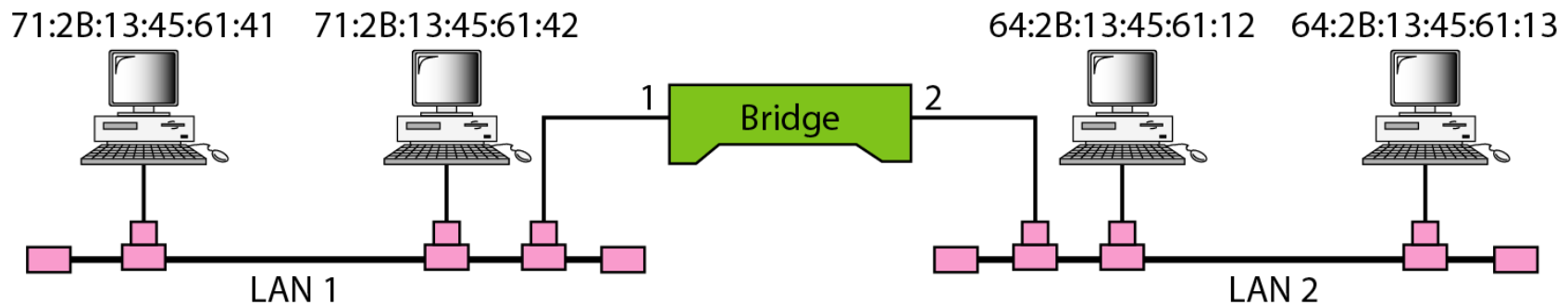
- A bridge operates in both the physical and the data link layer
 - As a physical layer device, it regenerates the signal it receives
 - As a data link layer device, the bridge can check the physical (MAC) addresses (source and destination) contained in the frame
- Bridge has **filtering** capability
 - It can check the destination address of a frame and decide if the frame should be forwarded or dropped
 - If the frame is to be forwarded, the decision must specify the port. A bridge has a table that maps addresses to ports.

A bridge connecting two LANs



Address	Port
71:2B:13:45:61:41	1
71:2B:13:45:61:42	1
64:2B:13:45:61:12	2
64:2B:13:45:61:13	2

Bridge Table





Transparent Bridge

- In a transparent bridge, the stations are completely unaware of the bridge's existence
- If a bridge is added or deleted from the system, reconfiguration of the stations is unnecessary
- Transparent bridges must meet three criteria:
 - Frames must be forwarded from one station to another.
 - The forwarding table is **automatically made by learning frame movements** in the network.
 - Loops in the system must be prevented.

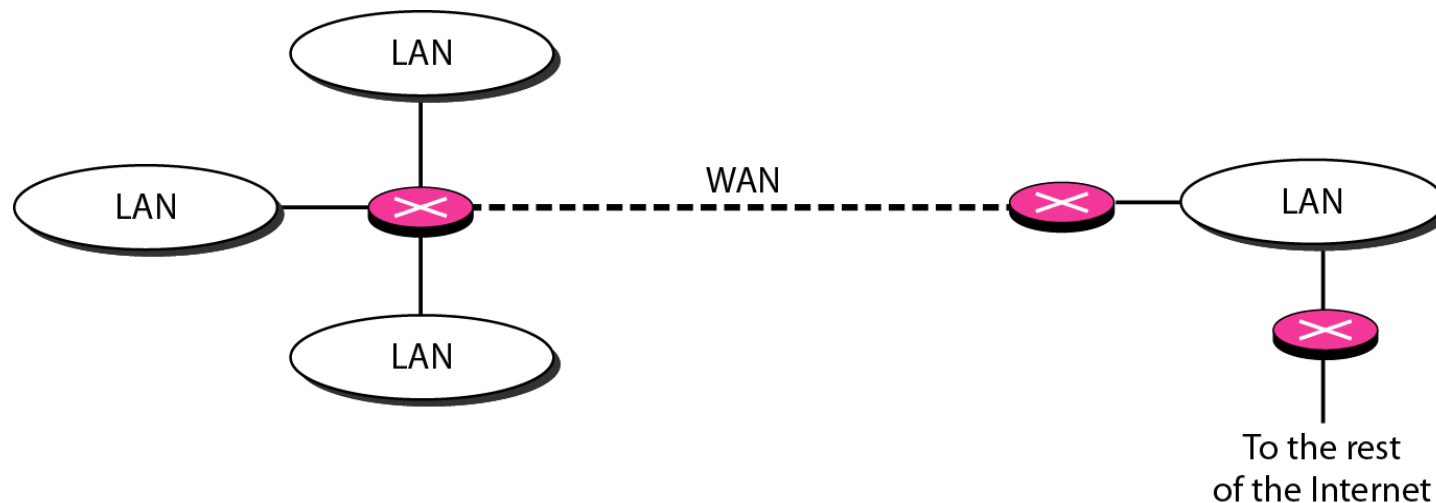


Two-Layer Switches

- Clarification
 - We can have a two-layer switch or a three-layer switch
 - A three-layer switch is used at the network layer (a kind of router)
 - The two-layer switch performs at the physical and data link layers (a bridge)
- A bridge with a few ports can connect a few LANs together
- A bridge with many ports may be able to allocate a unique port to each station
- A two-layer switch, as a bridge does, makes a filtering decision based on the MAC address of the frame it received
 - Can be more sophisticated
 - Can have a buffer to hold the frames for processing
 - Can have a switching factor that forwards the frames faster
 - Some new two-layer switches, called cut-through switches, have been designed to forward the frame as soon as they check the MAC addresses in the header of the frame.

Router

- A **router** is a **three-layer device** that routes packets based on their logical addresses (host-to-host addressing)
- Normally connects LANs and WANs in the Internet and has a routing table that is used for making decisions about the route
- The routing tables are normally dynamic and are updated using routing protocols
- A router, but a faster and more sophisticated. The switching fabric in a three-layer switch allows faster table lookup and forwarding





Gateway

- In certain scenarios Gateway and Routers are used interchangeably
- However, a gateway is normally a computer that operates in all five layers of the Internet or seven layers of OSI model
- A gateway takes an application message, reads it, and interprets it
- It can be used as a connecting device between two internetworks that use different models
- Gateways can provide security
- The gateway is used to filter unwanted application-layer messages.



BACKBONE NETWORKS

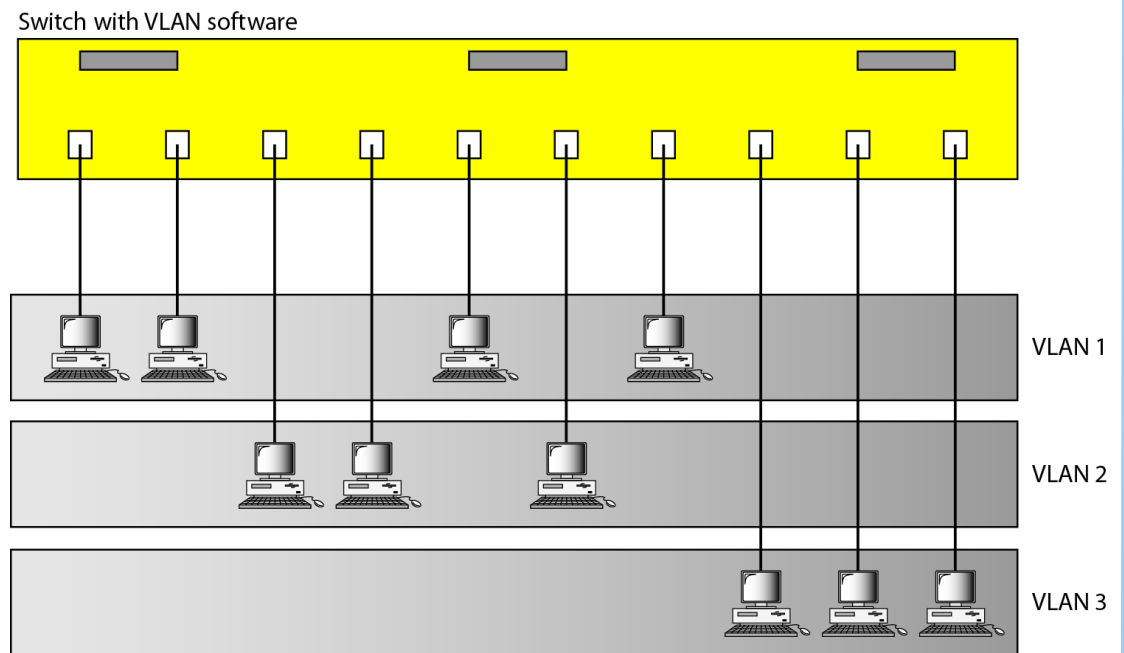
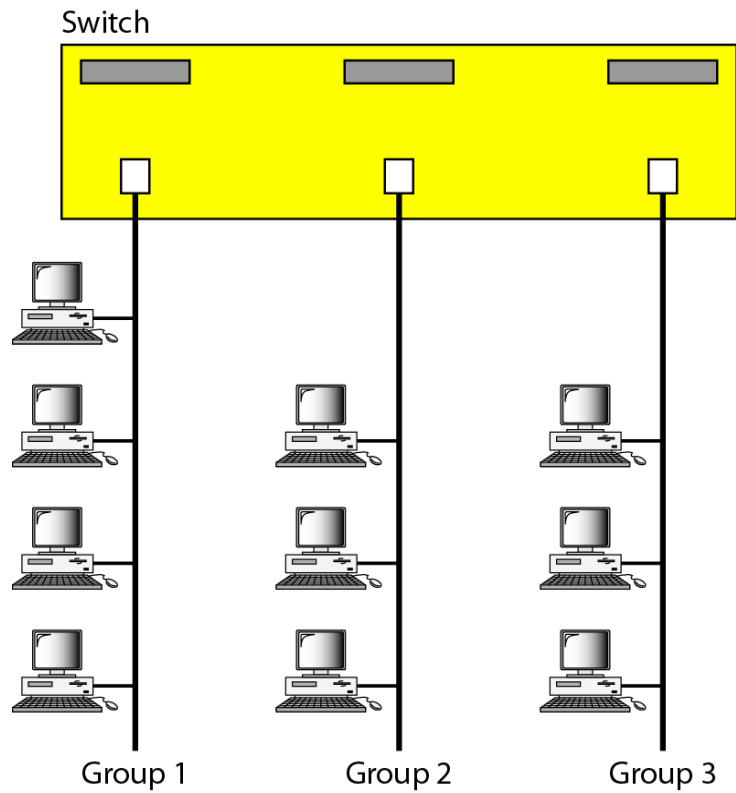
- Bus Backbone
- Star Backbone
- (left for self study)



VIRTUAL LANs

- Can roughly define a virtual local area network (VLAN) as a local area network configured by software, not by physical wiring
- Each VLAN is a work group in the organization
- If a person moves from one group to another, there is no need to change the physical configuration
- The group membership in VLANs is defined by software, not hardware
- All members belonging to a VLAN can receive broadcast messages sent to that particular VLAN
- The stations in a VLAN communicate with one another as though they belonged to a physical segment

LANs vs VLANs

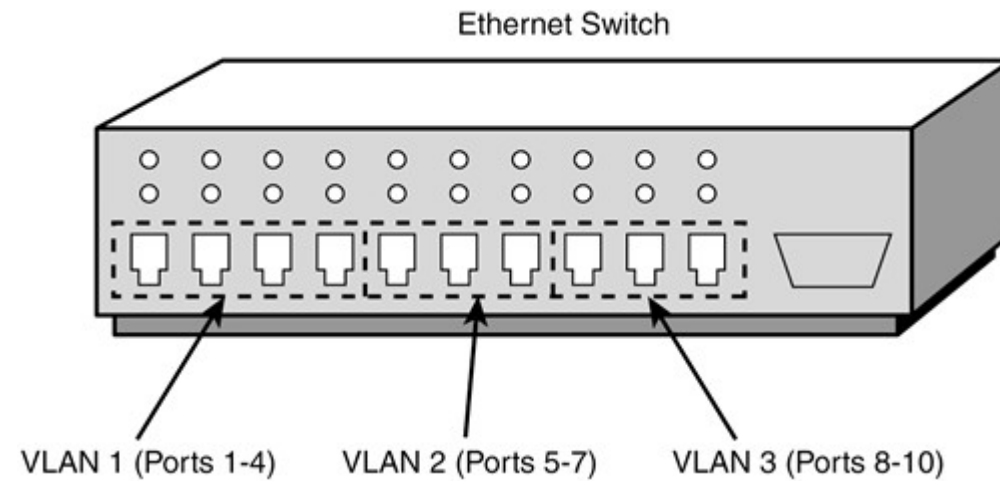




Membership in VLAN

- The characteristics that are commonly used to group the stations are
 - **Port Numbers**
 - For example, the administrator can define that stations connecting to ports 1, 2, 3, and 7 belong to VLAN 1; stations connecting to ports 4, 10, and 12 belong to VLAN 2; and so on.
 - **MAC Addresses**
 - Some VLAN vendors use the 48-bit MAC address as a membership characteristic
 - **IP Addresses**
 - For example, the administrator can stipulate that stations having IP addresses 181.34.23.67, 181.34.23.72, 181.34.23.98, and 181.34.23.112 belong to VLAN 1.
 - **Multicast IP Addresses**
 - Some VLAN vendors use the multicast IP address as a membership characteristic
 - **Combination**
 - The administrator can choose one or more characteristics when installing the software
- Membership Configuration can be **manual**, **automatic** or **semiautomatic**

Membership via port number





Communication Between Switches in VLANs

- Three methods have been devised for this purpose
- **Table Maintenance**
 - When a station sends a broadcast frame to its group members, the switch creates an entry in a table and records station membership
 - The switches send their tables to one another periodically for updating.
- **Frame Tagging**
 - An extra header is added to the MAC frame to define the destination VLAN
 - The frame tag is used by the receiving switches to determine the VLANs to be receiving the broadcast message.
- **Time-Division Multiplexing (TDM)**
 - In this method, the connection (trunk) between switches is divided into timeshared channels



Next : Network Layer